Meeting Minutes: NYU FRL Weekly Meeting

Saurabh and Ray – Friday 3rd April

Updates:

We focused on various discussions related to the Vague/ Complex route in this meeting. Ray showcased to me some of his progress on the complex route, which deals with the Memory route. We also discussed about a multimodal approach, where we "nudge" the MAS-XR toward a particular approach using reference images, discussed below

Vague/ Complex route:

The goal is to improve the Vague/ Complex route to interpret more accurately. We discussed literature reviews, which referenced novel spatial approaches using LLMs such as LayoutGPT, which says that sometimes a structural written prompt can help LLMs understand layouts much better than image recognition, owing to the fact that purely Transformer based (ViT) models might not be the best at object detection. This in turn becomes a Computer Vision problem, where we would have to explore objection detection, segmentation, semantic object detection etc., which is not in the best interest of our LLM focused research.

However, we agreed that LayoutGPT is an excellent proof of concept for your JSON based spatial input (for which we already see great results)

Image based Input to an LLM agent:

We proposed that we could use reference images of physical spaces such as Offices, Cafes, and Homes, as input to a separate agent, to help nudge the overall MAS-XR in the right direction. This means, we would have a strong Semantic Analyzer, as opposed to a purely spatial analysis.

<u>What is the Semantic approach in this system?</u>

Currently the system has five Agents, of which four are purely spatial analysts, and while the Language Agent does understand "intent" and "semantics", it's purely meant to analyze user command in natural language. That means to solve complex problems such as "turn my bedroom into a minimalist office room", we might need a Semantic Analyst Agent, that tries to understand the intent of the user based on a "aesthetic goal", (more visually!).

Image based Semantic Analysis:

Ray proposed that we can try to add an image into the system, which analyzes the space, such as adding an "OfficeView.png", with a vectorized floor plan view of all the objects in the scene, such as Sofa, Table, Chair, Desk etc. Saurabh proposed that instead of adding the image to the existing agents, and further adding scope for hallucinations, we make an entirely separate "Image Agent" that analyzes images, along with the user's intent.

Here are some of the things discussed in this section of the meeting:

1. We might have an Image Agent, purely meant for the vague/complex route, that gets fed the "target" scene, such as a Floor Plans of an office room/ Coffee Shop/ Bedroom etc.

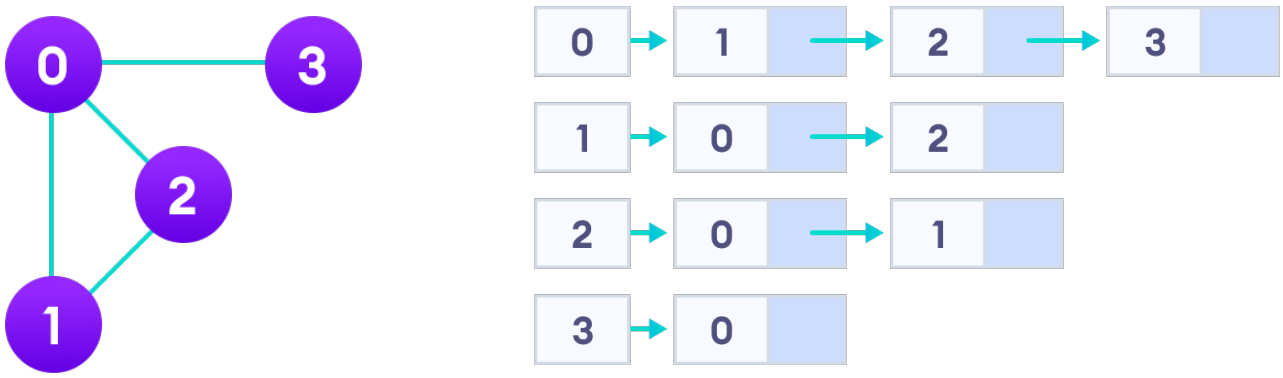
2. The Agent will analyze the various objects in the scene, and the spatial information related to each. E.g. Using a strict JSON format, the Agent might output things like

```
[
  {
    {"object-name" : "sofa"},
    {"spatial-info" : "sofa is to the right of the chair"}
  },
  {
    {add similar info for chair}
  }
]
```

3. For this reason, Ray thinks it is important to add well documented labels in the image itself, because the Image Agent might not understand what each object in the wireframe Floor Plan is (example: a simple rectangle would be a desk, but the LLM might not infer it by itself).
    a. For this, we might need to find clean and labeled images, with dimensions and each piece of furniture labeled correctly
4. We also discussed some alternate ways to represent spatial relationships between objects, specifically things such as Graph Adjacency Lists, and also SceneGraphs

Graph Adjacency Lists:

In classic Algorithms, a graph is represented as either an adjacency list or an Adjacency Matrix. Saurabh proposed, an adjacency list is as simple as making all the objects in the scene as "nodes" in the graph, and the adjacency list will have the objects relationship with all other objects (or the one's closer to it). Refer to following example derived from classic DSA:



In above example, if we represent "0", "1", "2", and "3" with the names for objects in the scene, the adjacency list may be written out in JSON as:

```
[
  {
    {“object-name” : “sofa”},
    {"adjacency" :
      {
        {"chair" : "right", "distance" : "1 meter"},
        {"table" : "left", "distance" : "2 meters"},
        {...more adjacency relationships}
      }
    }
  },

  {
    {add similar info for chair}
  }
]
```

Note: It is crucial to understand that while the representations are not novel, using an LLM to make decisions based on this structure might be novel. Further literature review studies would shed light on this proposal’s usefulness. It is also important to note that it will also significantly increase the LLM Agent Token usage with the Image and Spatial Graph based context input if this Semantic Route is ever implemented.

SceneGraph:

Due to Saurabh’s entry level experience with game engines, he proposed another way to represent the object hierarchy. A scene graph is a hierarchical tree data structure used in 3D engines and graphics software (like Unity or Open 3D Engine) to organize graphical scenes. Attached are some images that may help us understand SceneGraphs as a possible representation for our objects

Read the study here for more info made by RPI: ➔ LINK

The study showed an example which might help us understand how SceneGraphs work:

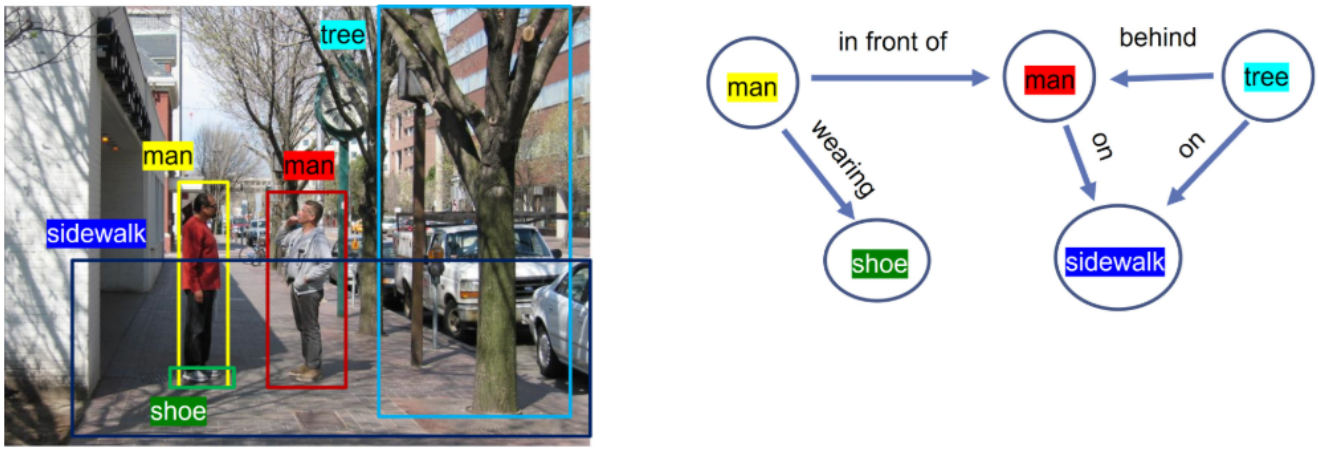


Fig. 1.1 - An image with different objects and its scene Graph

Limitations: Our scene might not have objects having a strict hierarchical relationship, so we would have to explore the feasibility of SceneGraphs in this projects.